

Why Enterprises Pay the Frontier Premium

Mechanisms behind the closed-vs-open procurement decision • modelled economics • **September version** • supersedes the August edition

THE ARGUMENT

Token price is the wrong denominator. Model spend runs **~1-2% of loaded labour** (structural, **HIGH**), and on threshold-gated work delivered value is convex rather than linear in capability (modelled, **MED**). Over long horizons* per-step reliability compounds – p^N is the upper bound on that decay, not a measurement of it, since real failures correlate and harnesses recover (section 1).

What changed since August is the open-weights ledger. Under this model's throughput assumption, self-hosting does not break even against a third-party open-weights API at any utilisation the fleet can reach – the escape hatch is **renting** open weights, not running them. That rests on 5,000 output tok/s per node, tagged **LOW** and the swing variable throughput. The genuine overpayment isn't choosing proprietary; it's **routing everything to a flagship max-effort tier**.

Correction of record. The August edition claimed a calibration advantage for the frontier tier; an interim draft of this version reversed it on newer hallucination rates. Both were wrong and both were this document's – the metric is conditional on non-correct answers and neither checked the denominator. Driver 4 is withdrawn, not flipped (section 5).

ABSTRACT – WHAT CHANGED SINCE THE AUGUST EDITION

Four load-bearing inputs moved. The thesis survives all four and reads stronger.

1 Reference prices were two generations stale

CORRECTED

The \$75/1M flagship line was legacy Opus 4.1 pricing. Frontier output now runs **\$25** (Opus 5) to **\$50** (Fable 5). Sonnet 5 held its introductory **\$2/\$10** as standard – the 1 September increase was cancelled. Every reference line in section 2 is redrawn.

2 Self-hosting lost its case entirely

REVISED

Hosted open-weights output fell from the assumed **\$6/1M** floor to a **~\$1.85 provider median**. Four nodes flat out for a year produce ~631B output tokens worth ~\$1.17M of API spend against \$1.81M fixed – **no utilisation at which the modelled fleet wins**. The mid-tier crossover moved to 28.7–34.3%, now inside the 20–35% real band.

3 The calibration driver was a denominator error

WITHDRAWN

The rate is **incorrect ÷ all non-correct responses**, so abstaining deflates it. Decomposed per 100 questions, absolute wrong assertions span only **20.1 to 27.9** across a lineup running 40–65% accuracy. Sonnet 5's "good" 39.4% yields **23.6 wrong answers** – more than Fable 5's "bad" 63.6% at **22.3**. The frontier tier is not less reliable; it attempts more and declines less.

4 The composite index caught up to p^N

REFRAMED

Intelligence Index v4.1 is roughly **50% agentic weight** (GDPval-AA v2 20%, Terminal-Bench 2.1 16%, τ^3 -Banking 14%), GDPval turn limit raised 100→250. The mechanism is unchanged; the claim that composites hide it is now only partly true.

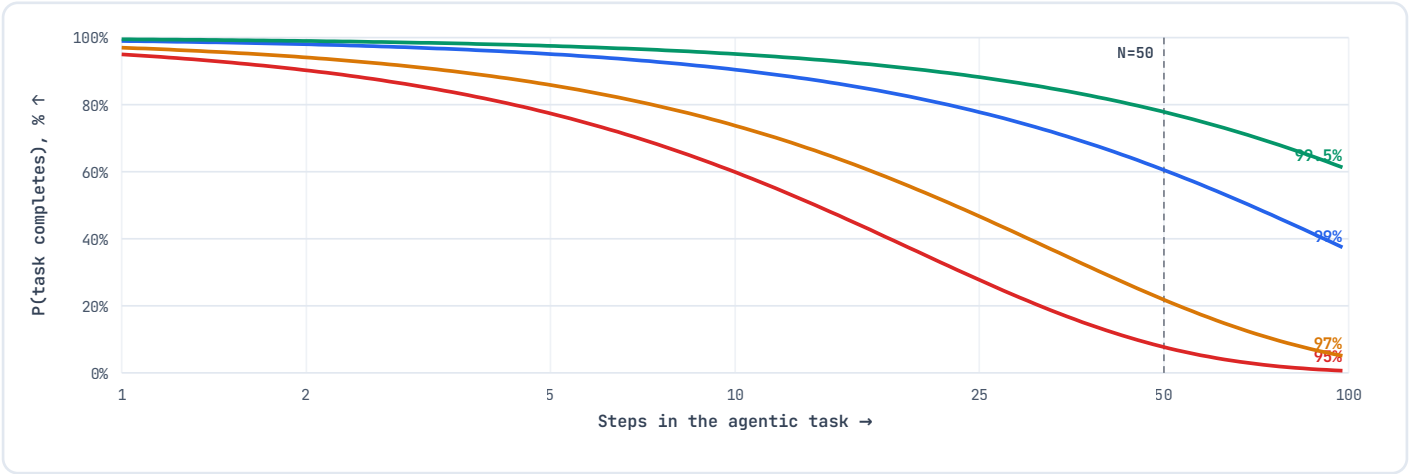
NET

The escape hatch from the frontier premium is **renting** open weights, not running them – the self-hosted quadrant has narrowed to a data-residency decision rather than an economic one. The overpayment that exists remains **failure to segment traffic**; the calibration finding does not add a second form, because absolute error is too flat across the lineup for model choice to be the lever. **Ground the fact-assertion work; route the long-horizon work by p^N .**

1 · Agentic error compounding

Task completion probability vs horizon · $P(\text{success}) \approx p^N$ · unchanged – derived arithmetic

The largest single driver. A two-point per-step reliability delta is immaterial on a single turn and decisive by step 50.



PER-STEP RELIABILITY	10 STEPS	25 STEPS	50 STEPS	100 STEPS
● 95.0%	59.9%	27.7%	7.7%	0.6%
● 97.0%	73.7%	46.7%	21.8%	4.8%
● 99.0%	90.4%	77.8%	60.5%	36.6%
● 99.5%	95.1%	88.2%	77.8%	60.6%

WHY THIS DOMINATES

Moving per-step reliability from 97% to 99% is a 2.8× swing at N=50 and 7.4× at N=100 – the boundary between "the agent ships the PR" and "an engineer babysits and re-drives." No token saving recovers a workflow that doesn't complete.

REVISED – WHAT THE MECHANISM DOES AND DOESN'T CLAIM

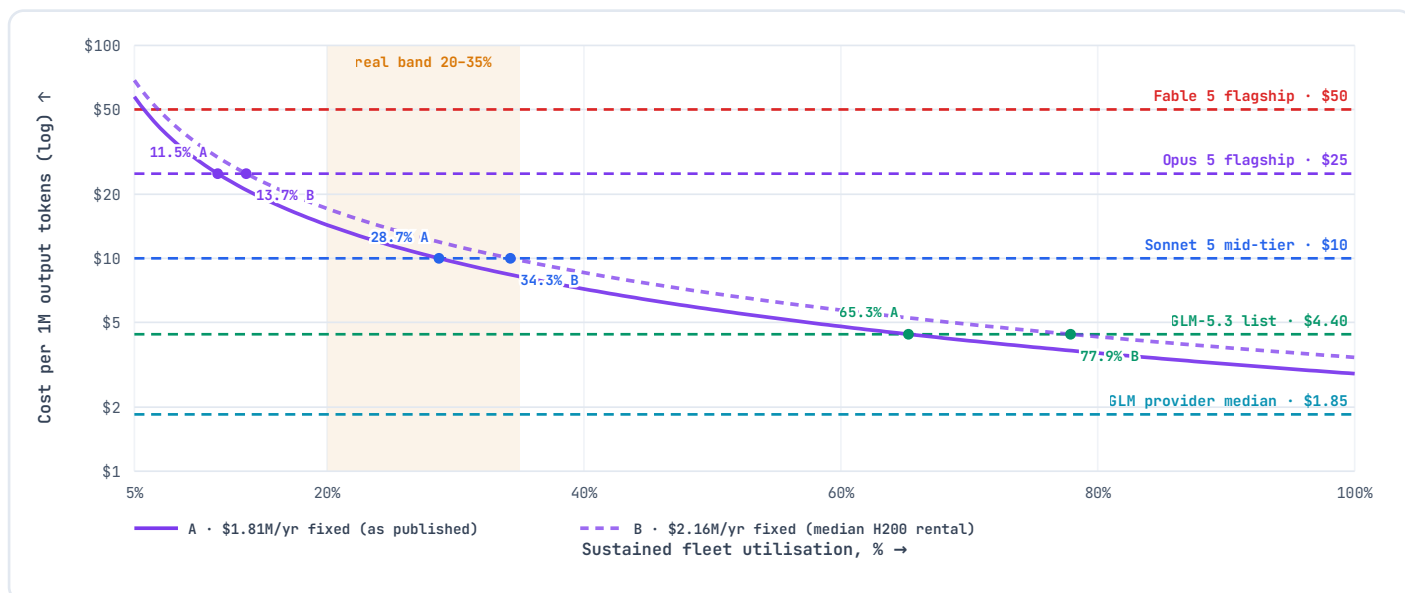
The benchmark caught up. The August edition called this mechanism invisible to composite indices. Intelligence Index v4.1 reweighted toward agentic workloads – GDPval-AA v2 at 20%, Terminal-Bench 2.1 at 16%, τ³-Bench Banking at 14%, roughly half the index – with the GDPval turn limit raised from 100 to 250 and IFBench removed for saturation. v4.1.1 is the current patch; Opus 5 holds #1 at Index 63. Composites now hide the mechanism less than they did three months ago.

And a limit on the curve above. p^N assumes independent Bernoulli steps. Real agent failures correlate – a bad plan poisons every subsequent step – and real harnesses recover, which is exactly what τ³-Banking's unhappy-path grading and the raised GDPval turn limit are built to measure. So the chart is an **upper bound on decay, not a forecast of completion**: the arithmetic is exact (HIGH) but its correspondence to observed agent behaviour is not (MED). The ordering survives – higher per-step reliability wins by a widening margin over horizon – but the specific 2.8× and 7.4× multipliers should be read as the shape of the effect, not as predicted completion rates for a given harness.

2 · Self-hosting total cost of ownership

Effective cost per 1M output tokens vs fleet utilisation · $4 \times 8 \times H200 + 3 \text{ ops FTE}$ · reference lines and fixed cost both revised

Open weights are free; serving them is not. Fixed cost amortises over tokens actually produced, so unit cost is governed by utilisation – and enterprise diurnal load runs 5-10× peak-to-trough. Two fixed-cost scenarios are now plotted: A retains the original \$26/hr node assumption, B uses the current market median.



BREAK-EVEN VS OPUS 5
FLAGSHIP \$25

11.5% / 13.7%
scenario A / B

BREAK-EVEN VS SONNET 5
MID-TIER \$10

28.7% / 34.3%
inside the real band

BREAK-EVEN VS GLM-5.3
LIST \$4.40

65.3% / 77.9%
above real load

BREAK-EVEN VS PROVIDER
MEDIAN \$1.85

never
155% / 185% util required

REVISED – THE CROSSOVER DISAPPEARED

The August edition put the open-weights crossover at 47.9% utilisation against a \$6/1M reference. That reference no longer exists in the market. GLM-5.2 lists at \$1.40/\$4.40 with a provider median nearer \$0.55/\$1.85, because open weights let third parties compete on hosting; DeepSeek V4 Pro sits at \$0.435/\$0.87, roughly 78% below Kimi K2.6 on output; DeepSeek V4 Flash is \$0.14/\$0.28. At a \$1.85 median, four nodes running flat out for a year produce ~631B output tokens worth ~\$1.17M of API spend against \$1.81M of fixed cost. **There is no utilisation at which the modelled fleet wins.** The mid-tier line moved the other way and is now the interesting one: 28.7% (A) to 34.3% (B) lands inside the 20-35% real band rather than below it, so self-hosting no longer cleanly beats mid-tier proprietary across real enterprise load – it straddles.

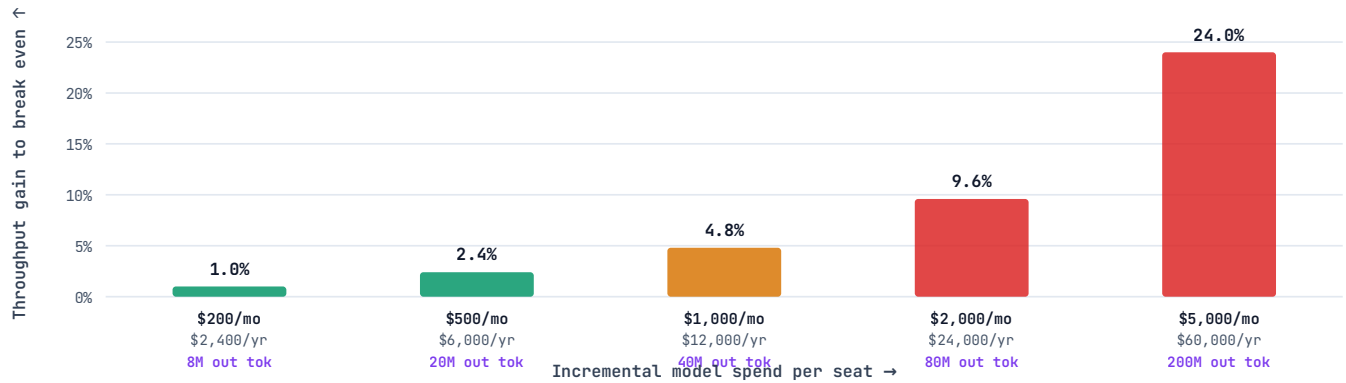
MODEL ASSUMPTIONS – LOAD-BEARING, TAGGED

Scenario A – infra \$911k/yr (4 nodes × \$26/hr – MED) plus ops \$900k/yr (3 FTE × \$300k loaded – MED) = **\$1.81M/yr fixed**, as originally published. **Scenario B** – H200 median on-demand is \$4.50/GPU-hr as of 28 August 2026, flat over 90 days, giving \$36/hr per 8-GPU node: infra \$1.26M/yr, **\$2.16M/yr fixed** (MED). The original \$26/hr sits near the verified floor (\$3.00/GPU-hr), not the median. Peak capacity of 5,000 output tok/s per node for a ~100B-class MoE under heavy batching remains **LOW** and is still the swing variable – halve throughput and every crossover doubles. The qualitative result survives that in one direction only: self-hosting beats a flagship price early, and beats a hosted open-weights API not at all.

3 • The labour denominator

Throughput gain required to repay the frontier premium • \$250k loaded labour per seat • arithmetic unchanged, threshold harder to reach

The premium is repaid in productivity, not tokens. The purple row is new: it shows the monthly output-token volume each spend level implies at current flagship pricing of \$25/1M.



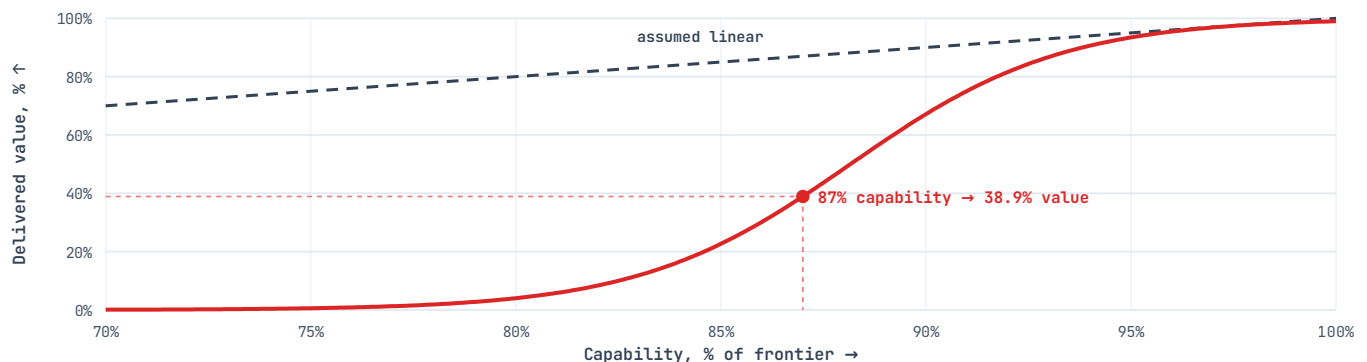
THE BAND WHERE THIS IS A NON-DECISION – NOW WIDER

At a realistic per-seat delta of **\$200-500/month**, break-even is a **1-2.4% throughput gain** – below measurement noise, which is exactly why enterprises stop optimising here. The argument only becomes live above ~\$2,000/seat/month, where a 9.6% lift is required. The flagship price cut from \$75 to \$25 per 1M output pushed that threshold further out: reaching \$2,000/seat/month now takes **~80M output tokens per seat per month**, against ~27M under the old reference. Sustained agentic work can still get there, but fewer seats do.

4 • Threshold-gated value

Delivered value vs capability • the composite-average trap, priced • illustrative

"Open weights reach ~87% of proprietary" assumes linear value transfer. On work gated by a pass threshold **delivered value is sigmoid**, and the last few index points sit on the knee.



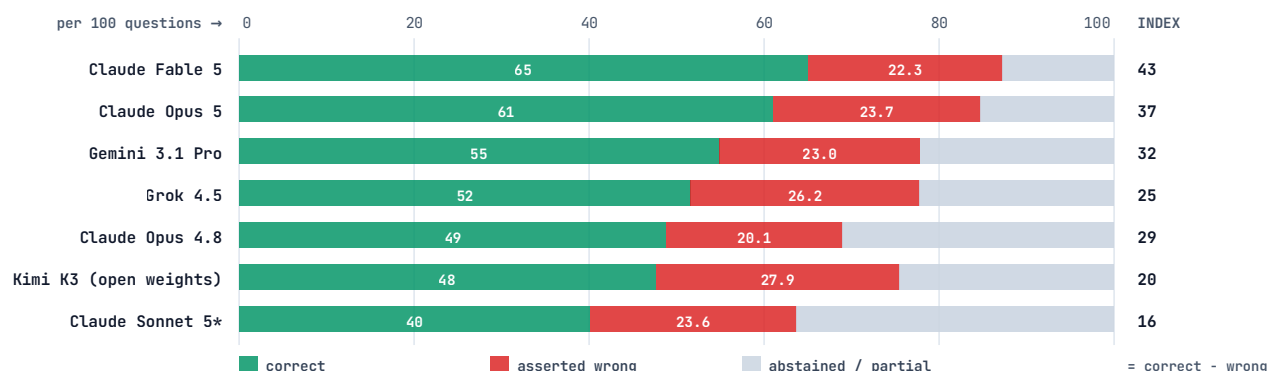
READING IT

On this curve 87% of frontier capability delivers ~39% of the value. The benchmark isn't wrong – a mean over nine or ten evals is simply the wrong functional form for threshold-gated work. The easy 80% of tasks were automatable two model generations ago; the residue is exactly what sits on the knee. **Caveat added this revision:** the 87% figure is the claim under test, not a verified current ratio. It came from the older index; no independently verified open-weights ratio against Intelligence Index v4.1.1 is in hand, and the v4.1 reweighting toward agentic tasks would likely move it. Treat the curve as **LOW** on curvature, **HIGH** on direction, and the 87% input as **MED** and aging.

5 • Calibration – what the rate actually measures

AA-Omniscience decomposed into correct / asserted-wrong / abstained per 100 questions • **rewritten – the August driver was wrong and so was its first correction**

The August edition cited a hallucination-rate gap as a frontier-premium driver. An interim draft of this version flipped that driver on newer rates. Both readings misuse the metric. AA defines hallucination rate as **incorrect ÷ (incorrect + partial + not attempted)** – a conditional share, not a share of output. Abstaining deflates it mechanically, so the rate cannot be compared across models that attempt different amounts. The decomposition below restores the denominator.



THE RESULT – ABSOLUTE ERROR IS NEARLY FLAT

Across a lineup spanning **40-65% accuracy** and a **1.6× spread in hallucination rate**, absolute wrong assertions vary only **1.4× – 20.1 to 27.9 per 100 questions**. Sonnet 5's flattering 39.4% rate produces **23.6 wrong assertions**, more than Fable 5's alarming 63.6% rate produces at **22.3** – because Sonnet 5 declines 36 of every 100 questions and Fable 5 declines 13. Fable 5 dominates Sonnet 5 on both axes at once: 25 points more accuracy and slightly fewer false assertions. The Index column, defined as correct minus incorrect, is the single number that already nets this out: Fable 5 at 43 against Sonnet 5 at 17.

WHAT THE ROUTING INSTRUCTION ACTUALLY IS

The frontier tier is not disqualified for anything. Hallucination rate is a closed-book recall property and says nothing about per-step agentic reliability or reasoning quality – those are Terminal-Bench 2.1, τ^3 -Banking, GDPval-AA v2 and GPQA/HLE/CritPt, which the frontier tier leads. Fable 5 tops AA-Briefcase for agentic knowledge work at 1,583 Elo against Sonnet 5 at 1,388 and Opus 5 at 1,354; Opus 5 leads Vals Index v2 at 67.2% and posts 96.0% on SWE-bench Verified. Three real instructions follow. **One:** Opus 4.8 is the genuine low-absolute-error pick at 20.1 wrong per 100 – but buys it with 16 points of accuracy, so it suits verification and checking, not research. **Two:** the fix for ungrounded fact assertion is grounding, not a model swap – the absolute-error band is too narrow for routing to move the needle. **Three:** a low rate bought by abstention is a trap; Cohere's Command A+ posts 14% hallucination alongside 9% accuracy, which is the reductio.

CONFIDENCE AND METHOD

Figures are the 26-27 August 2026 AA-synced snapshot. Every row reconciles against AA's published Index (Index = correct - incorrect) to within 0.2 points, which is why this set is used rather than the mixed-date compilation in the interim draft – **MED**, rising to **HIGH** for Fable 5 (65% / 43) and Opus 5 (61% / 37), confirmed directly on AA's page. ***Sonnet 5 accuracy of 40.1% is derived** from its published Index and rate, not read directly – **MED**. Models are excluded where only a rate was available and no denominator could be recovered: GPT-5.6 Sol (92.2%), GLM-5 (35.3%) and MiniMax M2.7 (35.6%) all appeared in the interim draft of this chart and are **withdrawn**, since a rate without an accuracy figure is uninterpretable and the open-weights comparison it supported was unsupported. A 2026 Nature paper (Kalai, Nachum, Vempala, Zhang) argues binary benchmark grading makes confident guessing the rational strategy, which is the structural reason this trade exists at all.

Net read

Eight drivers, with basis and confidence · changes flagged

DRIVER	MECHANISM	BASIS	CONF	Δ SEPT
Labor-dominated denominator	Model spend ≈ 1-2% of loaded engineer cost; flagship output cut to \$25/1M makes it more true, not less	Structural	HIGH	strengthened
Agentic error compounding	97%→99% per-step = 2.8× completion at N=50; now ~50% of Index v4.1 weight, so partly priced in	Derived (p ^N)	HIGH	reframed
Threshold-gated value	87% composite capability → ~39% delivered value on gated tasks; input ratio now aging	Modelled	MED	caveated
Calibration – withdrawn	Hallucination rate is conditional on non-correct answers, so abstention deflates it. Absolute wrong assertions span only 20.1-27.9 per 100 across the lineup – too narrow to support a premium or a routing rule. Ground the work instead	AA-Omniscience, 26-27 Aug 2026	MED	WITHDRAWN
Self-host TCO	No break-even against a \$1.85/1M hosted open-weights API at any utilisation; vs mid-tier at \$10 the crossover sits inside the 20-35% real band	Modelled	LOW	REVISED
Indemnity & compliance	IP indemnity, SOC 2 / HIPAA / FedRAMP, data residency	Contractual	HIGH	unchanged
Provenance & sanctions	Training-data lineage, backdoor risk, Entity List exposure	Qualitative	MED	unchanged
Harness, not checkpoint	Caching, batch, tool-call reliability, agentic scaffolds	Qualitative	HIGH	unchanged

LANE 1 · OPEN WEIGHTS

Bounded, high-volume, private Classification, extraction, routing, embedding. Error-tolerant and short-horizon. The revision narrows this lane's self-hosted form and widens its rented form: at \$0.28-\$1.85 per 1M output the hosted route wins on cost, and self-hosting is now a data-residency or air-gap decision rather than an economic one.

LANE 2 · MID-TIER PROPRIETARY

General reasoning, good-enough, well-calibrated
Where the dominated-flagship trap lives. Sonnet 5 held its \$2/\$10 introductory rate as standard pricing and scores 55 on AAIL v4.1.1. Note what it is not: a lower-error model. It asserts 23.6 wrong answers per 100 against Fable 5's 22.3, and only looks careful because it declines three times as often.

LANE 3 · FRONTIER

Long-horizon agentic, threshold-gated
Agentic chains and work sitting on the sigmoid knee. p^N is load-bearing and the token cost is the small line item. Fable 5 leads AA-Briefcase at 1,583 Elo and Opus 5 leads Vals Index v2 at 67.2%, so nothing in the calibration data narrows this lane. Ungrounded fact assertion is the one job to hand a retrieval layer rather than a bigger model.

NET

Enterprises don't pay the premium because they're bad at arithmetic – they pay it because the arithmetic they're doing has a different denominator. That holds. What the revision changes is where the open-weights win actually sits: not in self-hosting, which no longer clears a hosted API at any utilisation, but in renting open weights for bounded work at a fifth to a twentieth of mid-tier proprietary cost. The overpayment that exists is still failure to segment traffic. What the calibration work adds is a warning about the diagnosis rather than a new lane: **a hallucination rate compared across models that attempt different amounts is not a reliability measurement**, and reading one that way yields a routing rule that moves ~3 wrong answers per 100 while giving up 25 points of accuracy. Segment by horizon and by whether the work is grounded – not by a conditional rate.

Notes & clarifications

Plain-language expansion of a term used above

* WHAT p^N MEANS, AND WHAT IT DOESN'T

Take an agent on a 50-step task – read a file, edit it, run tests, read the output, and so on. If it gets each individual step right 97% of the time, the chance of getting through all 50 without a single mistake is 0.97 multiplied by itself 50 times, or about **22%**. Raise per-step reliability to 99% and it becomes **61%**. That multiplication is what p^N means, and it is why a two-point gap that looks trivial on a single question decides whether an agent finishes at all.

Two things make the real world kinder than that formula. First, the math assumes every step is an independent coin flip. Real agents aren't like that – if the model misreads the task at step 3 and forms a bad plan, steps 4 through 50 are not fresh chances to fail, they are all downstream of the same mistake. Failures cluster rather than scatter, so "50 independent opportunities to fail" overstates how many genuinely separate ones exist. Second, the math assumes one mistake ends the run. Real harnesses – the scaffolding around the model, with retries and error feedback – let the agent see a failed test, try something else, and carry on. A failed step often costs time rather than killing the task.

The recovery effect usually dominates, so observed completion rates come out **better** than the formula predicts. Hence "upper bound on the decay, not a measurement of it": read the curve in section 1 as the worst case and the shape of the problem, not a forecast. **What survives regardless is the ranking** – a more reliable model pulls further ahead the longer the task runs, and the gap widens with horizon. What should not travel into a business case is the specific multiplier: "2.8× better at 50 steps" describes the shape of the effect, not your harness's output.

Sources & confidence. Reliability compounding is derived arithmetic (p^N) – HIGH. Index composition, weights and rankings from Artificial Analysis Intelligence Index v4.1 and the v4.1.1 patch, read 30 August 2026 – HIGH. AA-Omniscience accuracy (Fable 5 65%, Opus 5 61%) from the live AA evaluation page – HIGH; remaining accuracy/rate pairs are the 26-27 Aug 2026 AA-synced snapshot, each reconciled against AA's published Index to within 0.2 points – MED. Sonnet 5 accuracy is derived from Index and rate – MED. AA-Briefcase Elo and Vals Index v2, Jul-Aug 2026 – MED. Anthropic list prices (Opus 5 \$5/\$25, Fable 5 \$10/\$50, Sonnet 5 \$2/\$10, Haiku 4.5 \$1/\$5) and the cancellation of the scheduled 1 September Sonnet 5 increase from Claude Platform pricing docs – HIGH. Open-weights pricing (GLM-5.2 \$1.40/\$4.40 list, ~\$0.55/\$1.85 provider median; GLM-5.3 \$1.40/\$4.40; DeepSeek V4 Pro \$0.435/\$0.87, V4 Flash \$0.14/\$0.28; Kimi K2.6 \$0.95/\$4.00, K3 \$3/\$15) from vendor pages and provider trackers, Jul-Aug 2026 – MED. H200 median on-demand \$4.50/GPU-hr across 29 providers, 28 Aug 2026, flat over 90 days – MED. Self-host TCO is a model, not a measurement: 5,000 output tok/s per node is an assumption (LOW) and dominates the result. Loaded labour \$250k is representative (MED). Threshold curve illustrative – LOW on curvature, HIGH on direction.

Not included. Reference prices are per 1M output tokens and exclude input, caching, and the verbosity tax – Claude 4.7 and later use a tokenizer producing roughly 30% more tokens for identical text, which shifts effective cost above these lines. Project-level discounts, committed-use pricing, enterprise volume agreements, Batch (50% off) and cache-hit (10% of base input) lanes are absent from every figure here and would compress the premium further.